

# Supplementary Information

Helium gas or liquid nitrogen for the first thermal shield of a 4 K cryostat: conjugate heat-transfer budget, coolant operating maps and work ledger

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All cases, scripts and tabulated results: [github.com/Imerson/Pre-cooling-a-superconducting-qubit-dilution-refrigerator](https://github.com/Imerson/Pre-cooling-a-superconducting-qubit-dilution-refrigerator) (release v2).

## S1 Enclosure model

**Table S1.** Regions, patches and boundary conditions of the enclosure CHT cases.

| Region                   | Patch                    | $T$ condition                              | Radiation condition                |
|--------------------------|--------------------------|--------------------------------------------|------------------------------------|
| domain0 (OVC-shield gap) | warmPlate, OVC           | fixed $T_{\text{hot}}$                     | grey diffuse, $\varepsilon = 0.04$ |
|                          | shield_slave             | fixed $T_{\text{cold}}$                    | grey diffuse, $\varepsilon = 0.02$ |
| domain1 (inner gap)      | warmPlate                | fixed $T_{\text{hot}}$                     | grey diffuse, $\varepsilon = 0.04$ |
|                          | coldPlate, shield_bottom | fixed $T_{\text{cold}}$                    | grey diffuse, $\varepsilon = 0.04$ |
|                          | shield                   | fixed $T_{\text{cold}}$                    | grey diffuse, $\varepsilon = 0.02$ |
|                          | domain1_to_coax_Lk       | conjugate (mapped)                         | grey diffuse, $\varepsilon = 0.02$ |
| coax_L1..L4 (solids)     | warm cap / cold cap      | fixed $T_{\text{hot}}$ / $T_{\text{cold}}$ | –                                  |
|                          | lateral                  | conjugate to domain1                       | –                                  |

**Table S2.** Gas-region settings that make the gaps behave as vacuum. Radiation in OpenFOAM’s **viewFactor** model is solved in fluid regions, so each gap is a fluid region made inert.

| Setting                       | Value                                                                          |
|-------------------------------|--------------------------------------------------------------------------------|
| pressure $p = p_{\text{rgh}}$ | $10^{-3}$ Pa ( <b>pRefValue</b> $10^{-3}$ )                                    |
| gravity                       | (0, 0, 0)                                                                      |
| transport                     | constant, $\mu = 10^{-11}$ Pa s, $\text{Pr} = 0.67$                            |
| density limits                | $\rho_{\text{max}} = 10^{-4}$ , $\rho_{\text{min}} = 10^{-12}$ kg m $^{-3}$    |
| radiation solver frequency    | every 10 iterations                                                            |
| agglomeration                 | <b>nFacesInCoarsestLevel</b> 250, <b>featureAngle</b> 20, <b>mergeLevels</b> 1 |

**Table S3.** Harness representation. Cable UT-085-SS-SS: 304 stainless outer conductor  $A = 1.585$  mm $^2$ , PTFE dielectric  $A = 2.001$  mm $^2$ , 304 stainless centre conductor  $A = 0.205$  mm $^2$ . Material  $k(T)$  from the NIST cryogenic fits  $\log_{10} k = \sum_n a_n (\log_{10} T)^n$ . Each modelled cylinder carries  $n_k$  lines with  $\kappa_{\text{eff}}(T) = (n_k/A_{\text{model}}) \sum_i A_i k_i(T)$ ;  $A_{\text{model}}$  is the calibrated effective conduction area of the 20 mm cylinder.

| Bundle  | Family  | $n_k$ | share of harness conduction |
|---------|---------|-------|-----------------------------|
| coax_L1 | drive   | 25    | 38.5 %                      |
| coax_L2 | flux    | 25    | 38.5 %                      |
| coax_L3 | readout | 10    | 15.4 %                      |
| coax_L4 | pump    | 5     | 7.7 %                       |

**Table S4.** Deployed  $\kappa_{\text{eff}}(T) = \sum_{n=0}^5 c_n T^n$  [W m<sup>-1</sup>K<sup>-1</sup>,  $T$  in K] per bundle and stage span (`kappaCoeffs<8>`, last two coefficients zero). Fit error < 0.5 % over each span.

| Span     | Bundle | $c_0$                    | $c_1$                   | $c_2$                    | $c_3$                    | $c_4$                     | $c_5$                     |
|----------|--------|--------------------------|-------------------------|--------------------------|--------------------------|---------------------------|---------------------------|
| 300→50 K | L1     | $-2.1522 \times 10^{-1}$ | $3.1627 \times 10^{-2}$ | $-2.3184 \times 10^{-4}$ | $9.9921 \times 10^{-7}$  | $-2.1908 \times 10^{-9}$  | $1.9383 \times 10^{-12}$  |
|          | L2     | $-2.1519 \times 10^{-1}$ | $3.1622 \times 10^{-2}$ | $-2.3181 \times 10^{-4}$ | $9.9907 \times 10^{-7}$  | $-2.1905 \times 10^{-9}$  | $1.9380 \times 10^{-12}$  |
|          | L3     | $-8.6092 \times 10^{-2}$ | $1.2651 \times 10^{-2}$ | $-9.2741 \times 10^{-5}$ | $3.9970 \times 10^{-7}$  | $-8.7636 \times 10^{-10}$ | $7.7535 \times 10^{-13}$  |
|          | L4     | $-4.3044 \times 10^{-2}$ | $6.3253 \times 10^{-3}$ | $-4.6369 \times 10^{-5}$ | $1.9984 \times 10^{-7}$  | $-4.3816 \times 10^{-10}$ | $3.8766 \times 10^{-13}$  |
| 300→77 K | L1     | $-1.3459 \times 10^{-1}$ | $2.9103 \times 10^{-2}$ | $-2.0205 \times 10^{-4}$ | $8.3171 \times 10^{-7}$  | $-1.7398 \times 10^{-9}$  | $1.4706 \times 10^{-12}$  |
|          | L2     | $-1.3457 \times 10^{-1}$ | $2.9099 \times 10^{-2}$ | $-2.0202 \times 10^{-4}$ | $8.3159 \times 10^{-7}$  | $-1.7396 \times 10^{-9}$  | $1.4704 \times 10^{-12}$  |
|          | L3     | $-5.3837 \times 10^{-2}$ | $1.1642 \times 10^{-2}$ | $-8.0824 \times 10^{-5}$ | $3.3269 \times 10^{-7}$  | $-6.9596 \times 10^{-10}$ | $5.8824 \times 10^{-13}$  |
|          | L4     | $-2.6917 \times 10^{-2}$ | $5.8206 \times 10^{-3}$ | $-4.0410 \times 10^{-5}$ | $1.6634 \times 10^{-7}$  | $-3.4796 \times 10^{-10}$ | $2.9411 \times 10^{-13}$  |
| 50→4 K   | L1     | $-4.8624 \times 10^{-4}$ | $9.1378 \times 10^{-4}$ | $3.6621 \times 10^{-5}$  | $-9.0343 \times 10^{-7}$ | $8.4798 \times 10^{-9}$   | $-2.8699 \times 10^{-11}$ |
|          | L2     | $-4.8624 \times 10^{-4}$ | $9.1378 \times 10^{-4}$ | $3.6621 \times 10^{-5}$  | $-9.0343 \times 10^{-7}$ | $8.4798 \times 10^{-9}$   | $-2.8699 \times 10^{-11}$ |
|          | L3     | $-1.9450 \times 10^{-4}$ | $3.6551 \times 10^{-4}$ | $1.4648 \times 10^{-5}$  | $-3.6137 \times 10^{-7}$ | $3.3919 \times 10^{-9}$   | $-1.1480 \times 10^{-11}$ |
|          | L4     | $-9.7247 \times 10^{-5}$ | $1.8275 \times 10^{-4}$ | $7.3242 \times 10^{-6}$  | $-1.8068 \times 10^{-7}$ | $1.6959 \times 10^{-9}$   | $-5.7398 \times 10^{-12}$ |
| 77→4 K   | L1     | $-7.4665 \times 10^{-4}$ | $1.2814 \times 10^{-3}$ | $4.9472 \times 10^{-5}$  | $-1.2245 \times 10^{-6}$ | $1.1714 \times 10^{-8}$   | $-4.2617 \times 10^{-11}$ |
|          | L2     | $-7.4665 \times 10^{-4}$ | $1.2814 \times 10^{-3}$ | $4.9472 \times 10^{-5}$  | $-1.2245 \times 10^{-6}$ | $1.1714 \times 10^{-8}$   | $-4.2617 \times 10^{-11}$ |
|          | L3     | $-2.9865 \times 10^{-4}$ | $5.1254 \times 10^{-4}$ | $1.9788 \times 10^{-5}$  | $-4.8978 \times 10^{-7}$ | $4.6855 \times 10^{-9}$   | $-1.7046 \times 10^{-11}$ |
|          | L4     | $-1.4932 \times 10^{-4}$ | $2.5626 \times 10^{-4}$ | $9.8938 \times 10^{-6}$  | $-2.4488 \times 10^{-7}$ | $2.3427 \times 10^{-9}$   | $-8.5230 \times 10^{-12}$ |

**Free-molecular gas conduction bound.** At  $p < 10^{-4}$  mbar the mean free path exceeds the gap and gas heat transport is free-molecular,  $\dot{Q}_{\text{gas}} = \alpha \Lambda p A \Delta T$  [1]; it is linear in pressure and below 1 % of the harness conduction at the operating pressure, so it is carried as a budget line.

**Table S5.** View-factor agglomeration refinement and row normalisation. Closure = mean row sum  $\sum_j F_{ij}$  of the generated matrix (unity for a closed enclosure); loads with the matrix as generated and, at the production level, with rows normalised.

| Case     | Coarsest-level faces | Closure (mean; range) | Radiation, as generated | Radiation, normalised         |
|----------|----------------------|-----------------------|-------------------------|-------------------------------|
| 300→50 K | 60                   | –                     | 9.46 W                  | –                             |
| 300→50 K | 250 (production)     | 0.99–1.01; 0.50–1.13  | 8.41 W                  | 7.36 W                        |
| 300→50 K | 400                  | 0.98–0.99             | 6.49 W                  | –                             |
| 300→50 K | 800                  | –                     | 3.26 W                  | –                             |
| 300→50 K | closed form          | –                     | 7.07 W                  | –                             |
| 300→77 K | 250 (production)     | 0.99                  | 8.38 W                  | 7.34 W (closed form 7.04 W)   |
| 50→4 K   | 250 (production)     | 1.00                  | 5.29 mW                 | 3.16 mW (closed form 3.15 mW) |
| 77→4 K   | 250 (production)     | 1.00                  | 29.7 mW                 | 17.8 mW (closed form 17.7 mW) |

**Row normalisation.** Each fluid region of the enclosure model is a closed surface, so every row of the view-factor matrix must sum to unity. The generated matrix (`viewFactorsGen`, Gauss quadrature tolerance 0.01) is stored sparsely, one entry per visible face pair; the normalisation divides each row by its sum and leaves rows with a sum below 0.2 (none occurred) untouched. The solve was then restarted from the converged production field for 600 further iterations. Script `normalise_F.py` in the repository.

**Table S6.** Emissivity sensitivity (production view factors) and MLI-equivalent variant.

| Case               | $\varepsilon$ warm / cold | Radiation [W] | Total [W] |
|--------------------|---------------------------|---------------|-----------|
| 300→50 K, low      | 0.02 / 0.01               | 5.45          | 6.48      |
| 300→50 K, baseline | 0.04 / 0.02               | 8.41          | 9.44      |
| 300→50 K, high     | 0.06 / 0.03               | 11.48         | 12.51     |
| 300→77 K, low      | 0.02 / 0.01               | 5.44          | 6.40      |
| 300→77 K, baseline | 0.04 / 0.02               | 8.38          | 9.35      |
| 300→77 K, high     | 0.06 / 0.03               | 11.44         | 12.40     |
| 300→50 K, MLI-eq.  | 0.04 / 0.003              | 1.81          | 2.84      |
| 300→77 K, MLI-eq.  | 0.04 / 0.003              | 1.81          | 2.77      |

## S2 Cold-plate sweep: full results

**Table S7.** All cold-plate runs at 9.4 W (production mesh).  $\dot{m}$  in  $\text{g s}^{-1}$ , volumetric flow in  $\text{cm}^3 \text{s}^{-1}$ ,  $\Delta p$  in Pa,  $\dot{W}_p$  in  $\mu\text{W}$ , temperatures in K,  $h$  in  $\text{W m}^{-2}\text{K}^{-1}$ ;  $T_{\text{in}} = 45 \text{ K}$  (He), 77 K (LN<sub>2</sub>); saturation margin at 3 bar;  $y^+$  = maximum on the heated wall; closure = energy-balance error. Gravity-on (g on) nitrogen runs did not reach a steady state (see closure).

| Case                                   | $Re$  | $\dot{m}$ | $\dot{V}$ | $\Delta p$ | $\dot{W}_p$ | $T_{\text{out}}$ | $T_w$ | $T_{w,\text{max}}$ | $T_w - T_b$ | $h$  | $Nu$ | $f$    | NTU   | $\varepsilon$ | Gr/ $Re^2$ | margin | $y^+$ | closure % |
|----------------------------------------|-------|-----------|-----------|------------|-------------|------------------|-------|--------------------|-------------|------|------|--------|-------|---------------|------------|--------|-------|-----------|
| He 1 bar $Re=500$                      | 500   | 0.159     | 167       | 0.294      | 48.6        | 56.36            | 87.4  | 97.6               | 36.72       | 17.1 | 3.66 | 0.186  | 0.310 | 0.266         | 0.63       | –      | –     | -0.06     |
| He 1 bar $Re=500$ (g on)               | 500   | 0.159     | 167       | 0.331      | 54.6        | 56.21            | 67.5  | 48.4               | 16.87       | 37.1 | 7.96 | 0.21   | 0.674 | 0.490         | 0.29       | –      | –     | -1.41     |
| He 1 bar $Re=1000$                     | 1000  | 0.318     | 316       | 0.671      | 211         | 50.68            | 76.8  | 83.0               | 28.94       | 21.7 | 4.64 | 0.112  | 0.196 | 0.178         | 0.15       | –      | –     | -0.05     |
| He 1 bar $Re=1500$                     | 1500  | 0.477     | 465       | 1.12       | 521         | 48.79            | 71.7  | 75.8               | 24.77       | 25.3 | 5.42 | 0.0847 | 0.153 | 0.142         | 0.06       | –      | –     | -0.05     |
| He 1 bar $Re=2300$                     | 2300  | 0.731     | 702       | 1.98       | 1.39e+03    | 47.47            | 67.1  | 69.4               | 20.82       | 30.1 | 6.45 | 0.0642 | 0.119 | 0.112         | 0.022      | –      | –     | -0.06     |
| He 1 bar $Re=2300$ (g on)              | 2300  | 0.731     | 703       | 2.02       | 1.42e+03    | 47.48            | 62.7  | 59.3               | 16.41       | 38.2 | 8.18 | 0.0656 | 0.151 | 0.140         | 0.018      | –      | –     | 0.55      |
| He 1 bar $Re=5000$                     | 5000  | 1.59      | 1.5e+03   | 6.62       | 9.95e+03    | 46.14            | 55.3  | 51.9               | 9.73        | 64.4 | 13.8 | 0.0461 | 0.117 | 0.110         | 0.0023     | –      | 2.9   | -0.08     |
| He 1 bar $Re=10000$                    | 10000 | 3.18      | 2.99e+03  | 21.8       | 6.51e+04    | 45.57            | 50.3  | 47.9               | 5.00        | 125  | 26.8 | 0.0382 | 0.114 | 0.107         | 0.0003     | –      | 5.6   | -0.18     |
| He 5 bar $Re=2300$                     | 2300  | 0.731     | 140       | 0.395      | 55.5        | 47.47            | 67.1  | 69.4               | 20.82       | 30.1 | 6.45 | 0.0641 | 0.119 | 0.112         | 0.56       | –      | –     | -0.04     |
| He 18 bar $Re=2300$                    | 2300  | 0.731     | 39        | 0.11       | 4.29        | 47.47            | 67.1  | 69.4               | 20.82       | 30.1 | 6.45 | 0.0643 | 0.119 | 0.112         | 7.3        | –      | –     | -0.04     |
| LN <sub>2</sub> 3 bar $Re=500$         | 500   | 4.09      | 5.06      | 0.202      | 1.02        | 78.13            | 85.0  | 85.7               | 7.46        | 84   | 5.77 | 0.163  | 0.151 | 0.140         | 40         | 2.2    | –     | 0.00      |
| LN <sub>2</sub> 3 bar $Re=500$ (g on)  | 500   | 4.09      | 5.06      | 1.68       | 8.5         | 77.32            | 78.9  | 78.6               | 1.74        | 361  | 24.8 | 1.35   | 0.650 | 0.478         | 9.4        | 9.3    | –     | -71.96    |
| LN <sub>2</sub> 3 bar $Re=1000$        | 1000  | 8.17      | 10.1      | 0.501      | 5.07        | 77.56            | 83.1  | 83.1               | 5.77        | 109  | 7.45 | 0.101  | 0.098 | 0.093         | 7.8        | 4.8    | –     | 0.00      |
| LN <sub>2</sub> 3 bar $Re=1500$        | 1500  | 12.3      | 15.2      | 0.869      | 13.2        | 77.38            | 82.1  | 81.9               | 4.92        | 127  | 8.75 | 0.0778 | 0.076 | 0.074         | 2.9        | 6.0    | –     | 0.00      |
| LN <sub>2</sub> 3 bar $Re=2300$        | 2300  | 18.8      | 23.3      | 1.57       | 36.6        | 77.25            | 81.2  | 80.8               | 4.12        | 152  | 10.4 | 0.0599 | 0.060 | 0.058         | 1          | 7.1    | –     | -0.00     |
| LN <sub>2</sub> 3 bar $Re=2300$ (g on) | 2300  | 18.8      | 23.3      | 3          | 69.8        | 77.22            | 78.5  | 77.6               | 1.35        | 463  | 31.8 | 0.114  | 0.181 | 0.166         | 0.34       | 10.4   | –     | -10.77    |
| LN <sub>2</sub> 3 bar $Re=5000$        | 5000  | 40.9      | 50.6      | 5.57       | 282         | 77.11            | 78.6  | 77.6               | 1.55        | 405  | 27.8 | 0.0449 | 0.073 | 0.070         | 0.083      | 10.3   | 2.9   | 0.04      |
| LN <sub>2</sub> 3 bar $Re=10000$       | 10000 | 81.7      | 101       | 18.6       | 1.88e+03    | 77.06            | 77.8  | 77.3               | 0.81        | 777  | 53.3 | 0.0374 | 0.070 | 0.068         | 0.011      | 10.6   | 5.6   | 0.03      |

**Table S8.** Grid convergence index study for both fluids at  $Re = 2300$  (Celik et al. 2008); meshes 86 400 / 291 600 / 984 150 cells, refinement ratio 1.5, production mesh = medium.

| Fluid                   | Quantity                               | Coarse | Medium | Fine   | $p$  | Richardson | GCI <sub>medium</sub> | GCI <sub>fine</sub> |
|-------------------------|----------------------------------------|--------|--------|--------|------|------------|-----------------------|---------------------|
| He, 1 bar               | $Nu$                                   | 6.439  | 6.449  | 6.457  | 0.64 | 6.485      | 0.70 %                | 0.54 %              |
| He, 1 bar               | $f$                                    | 0.0632 | 0.0642 | 0.0647 | 1.40 | 0.0655     | 2.5 %                 | 1.4 %               |
| He, 1 bar               | $h$ [ $\text{W m}^{-2}\text{K}^{-1}$ ] | 30.05  | 30.10  | 30.14  | 0.64 | 30.27      | 0.70 %                | 0.54 %              |
| LN <sub>2</sub> , 3 bar | $Nu$                                   | 10.513 | 10.436 | 10.382 | 0.87 | 10.26      | 2.2 %                 | 1.5 %               |
| LN <sub>2</sub> , 3 bar | $f$                                    | 0.0592 | 0.0599 | 0.0603 | 1.71 | 0.0607     | 1.6 %                 | 0.8 %               |
| LN <sub>2</sub> , 3 bar | $h$ [ $\text{W m}^{-2}\text{K}^{-1}$ ] | 153.2  | 152.1  | 151.3  | 0.87 | 149.5      | 2.2 %                 | 1.5 %               |

**Property correction and energy accounting.** The helium transport properties of the earlier version of this comparison ( $\mu = 2.73 \mu\text{Pa s}$ ,  $k = 0.0185 \text{ W m}^{-1}\text{K}^{-1}$ ) correspond to helium near 15 K; the present runs use the 50 K values ( $6.36 \mu\text{Pa s}$ ,  $0.0467 \text{ W m}^{-1}\text{K}^{-1}$ , CoolProp/NIST). The mixing-cup outlet temperature is the mass-flux-weighted average  $\sum \phi_f T_f / \sum \phi_f$  over the outlet faces (surfaceFieldValue, weightedAverage with phi); the weightedAreaAverage operation, which multiplies the mass flux by the face area a second time, biases the outlet temperature low on a graded mesh and was the cause of the 2–10 % energy imbalances reported earlier. The delivered wall heat flux was verified with the solver-native wallHeatFlux function object (9.395 W).

## S3 Hardware efficiencies used in the real ledger

**Table S9.** Stage efficiencies as fractions of Carnot.

| Stage                    | $\eta$ range | Source                                                                                                                                                                                                        |
|--------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 K cryocooler stage     | 0.008–0.015  | PT415: 1.5 W at 4.2 K with 40 W at 45 K for 10.7 kW [2]; input allocated by Carnot work gives 0.031 for both stages combined; Radebaugh: “at 4 K efficiencies of about 1 % of Carnot or less are typical” [3] |
| He first stage (45–80 K) | 0.10–0.15    | Radebaugh: 10–15 % of Carnot for the best commercial 80 K coolers; large plants up to 30 % [3, 4]                                                                                                             |
| LN <sub>2</sub> supply   | 0.30–0.40    | ideal liquefaction work 0.213 kWh kg <sup>-1</sup> [5]; plant consumption 0.5–0.7 kWh kg <sup>-1</sup> [6]; latent heat 199 kJ kg <sup>-1</sup>                                                               |

## References

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